

Annotated Bibliography

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February 17, 2026

References

- [1] Alex Chao, Jeffrey M. Minucci, Troy M. Ferland, E. Tyler Carr, Greg Janesch, Safia Rizwan, Heather D. Whitehead, Tommy Cathey, Shirley Pu, Laura D. Brunelle, Angela L. Batt, Jon R. Sobus, and Antony J. Williams. Prioritizing chemical candidates from non-targeted analysis using metadata, spectral similarity, and hazard scoring within interpret nta. *Analytical Chemistry*, 97(29):15904–15912, 2025. PMID: 40670324.

This study introduces INTERPRET NTA, a computational tool that enhances non-targeted analysis (NTA) by integrating spectral similarity, metadata, and hazard data to prioritize chemical candidates detected via high-resolution mass spectrometry (HRMS). While traditional NTA approaches focus on structure annotation based primarily on mass spectral matching, this work emphasizes data-driven prioritization and visualization strategies that facilitate the identification of relevant chemicals within complex environmental matrices. The methodologies described, particularly the use of spectral similarity scoring and interactive visualization, align with advanced computational approaches such as molecular networking and cluster analysis. These techniques are critical for exploring unknown molecules in HRMS datasets. Applying these strategies to remote environmental samples like those from Mount Everest, Kilimanjaro, or Tibetan rivers could significantly enhance the discovery and characterization of unannotated organic contaminants by enabling network-based clustering and contextual prioritization beyond mere structure assignment. This work informs the development of computational frameworks that support the spectral and metadata-driven exploration of large and complex HRMS datasets, which is highly relevant for research efforts aiming to discover and annotate unknown environmental molecules in remote terrestrial and ecological contexts.

- [2] Madeleine Ernst, Kyo Bin Kang, Andrés Mauricio Caraballo-Rodríguez, Louis-Felix Nothias, Joe Wandy, Christopher Chen, Mingxun Wang, Simon Rogers, Marnix H. Medema, Pieter C. Dorrestein, and Justin J.J. van der Hooft. Molnetenhancer: Enhanced molecular networks by integrating metabolome mining and annotation tools. *Metabolites*, 9(7), 2019.

This paper on MolNetEnhancer aligns closely with my research focus on the computational identification of unknown molecules using high-resolution mass spectrometry. While traditional approaches often rely solely on structure annotation for unknowns, MolNetEnhancer exemplifies an advanced, integrative methodology that leverages molecular networking, substructure discovery (MS2LDA), and in silico annotation tools to explore the chemical diversity within complex datasets. Its emphasis on non-targeted, cluster-based analysis for natural and synthetic metabolomes provides a valuable framework for my work, particularly in analyzing organic contaminants or complex mixtures from remote areas such as Mount Everest, Mt. Kilimanjaro, and Tibet River. By employing similar network-based strategies, I can enhance the detection, classification, and structural inference of unknown molecules beyond mere mass data, facilitating deeper insights into uncharacterized environmental chemicals through computational metabolomics.

- [3] Wenkai Fan. Suspect screening and semi-quantification of organic contaminants in snow and water samples on the qinghai–tibet plateau and high mountain regions. Undergraduate thesis, The College of Wooster, Wooster, OH, 2025. Unpublished thesis.

This thesis provides a comprehensive application of suspect screening and semi-quantitative analysis of organic contaminants in remote high-altitude environments using high-resolution mass spectrometry (HRMS). While the primary focus is on environmental chemistry and pollution transport mechanisms, it also exemplifies the use of advanced analytical techniques and data interpretation methods relevant to computational approaches for molecule identification. The utilization of HRMS data to detect and tentatively identify a broad spectrum of known contaminants aligns with the goals of non-targeted analysis in metabolomics and environmental studies. Furthermore, the study’s emphasis on suspect screening, combined with high-resolution data acquisition, underscores the importance of sophisticated data mining strategies—such as molecular networking, clustering, and multidimensional analysis—in revealing unknown or unexpected compounds. As such, this work offers valuable insights and a real-world context for developing computational frameworks aimed at the annotation and structural elucidation of unknown molecules, especially when integrated with molecular networking tools and machine learning algorithms in cheminformatics.

- [4] Peter K.T. Newton-S.R. et al. Fisher, C.M. Approaches for assessing performance of high-resolution mass spectrometry–based non-targeted analysis methods. *Anal Bioanal Chem*, 414:6455–6471, 2022.

This research paper provides a comprehensive overview of the current state and challenges associated with performance assessment of high-resolution mass spectrometry (HRMS)–based non-targeted analysis (NTA) methods, emphasizing the need for standardized metrics and frameworks. For my study on computational identification of unknown molecules via HRMS, particularly

in analyzing organic contaminants in remote environments such as Mount Everest, Mt. Kilimanjaro, and Tibetan rivers, this paper offers valuable insights into the evaluation criteria and reliability of NTA workflows beyond simple structural assignment. It highlights the importance of robust performance metrics, which are critical when deploying computational approaches like molecular networking and cluster analysis to interpret complex environmental data. The focus on qualitative performance metrics such as confidence in chemical identification can guide my development of algorithms for spectral clustering and molecular network construction. These strategies will ensure meaningful differentiation of unknowns and proper assessment of confidence levels. Overall, the paper situates my computational methods within the broader context of NTA validity and underscores the necessity of standardized evaluation strategies. These strategies are especially relevant when working with large, diverse datasets collected from remote or extreme locations.

- [5] Tobias Kind, Hiroshi Tsugawa, Tomas Cajka, Yan Ma, Zijuan Lai, Sajjan S. Mehta, Gert Wohlgemuth, Dinesh Kumar Barupal, Megan R. Showalter, Masanori Arita, and Oliver Fiehn. Identification of small molecules using accurate mass ms/ms search. *Mass Spectrometry Reviews*, 37(4):513–532, 2018.

This review explains how to identify small molecules using high-resolution MS/MS techniques. It highlights the importance of spectral libraries that include experimental data and computer-generated spectra. The article covers methods for creating in silico spectra using various software tools and developing new databases. It also stresses the need for proper data curation and quality control. From my point of view, it provides valuable insights into how computational approaches and shared spectral resources can aid in analyzing complex samples. These methods are especially helpful for identifying unknown compounds in environmental or natural product research.

- [6] Feerick A.-Braun J. M. Feng Y. L. Hall A. Koelmel J. Manzano C. Newton S. R. Pennell K. D. Place B. J. Godri Pollitt K. J. Prasse C. Young J. A. Manz, K. E. Non-targeted analysis (nta) and suspect screening analysis (ssa): A review of examining the chemical exposome. *Journal of Exposure Science Environmental Epidemiology*, 33(4):524–536, 2023.

This review article critically discusses the application of high-resolution mass spectrometry (HRMS) in non-targeted analysis (NTA) and suspect screening analysis (SSA) for environmental and human exposure assessment. It highlights the importance of computational tools, database searching, and software workflows in identifying unknown chemicals within complex matrices. Although traditional NTA focuses on assigning structures based on spectral data, this paper underscores emerging strategies such as spectral library matching, in silico predictions, and the use of software for molecular

formula generation. Importantly, it identifies gaps in open-source software development and emphasizes the need for advanced computational approaches such as molecular networking and cluster analysis to enhance the annotation and interpretation of unidentified HRMS features. This aligns with my research focus on employing computational techniques to facilitate the molecular networking of unknown compounds detected in remote environmental samples. It emphasizes moving beyond mere mass-based identification towards relationship-based clustering and annotation of molecular families. The review’s discussion on integrating multiple analytical platforms and expanding spectral libraries directly informs the development of machine learning and network-based tools for the comprehensive characterization of organic contaminants in extreme and understudied environments.

- [7] Stephanie Nino-Suastegui, Eve Painter, Jameson W. Sprankle, Jillian J. Morrison, Jennifer A. Faust, and Rebekah Gray. Non-targeted analysis and suspect screening of organic contaminants in temperate snowfall using liquid chromatography high-resolution mass spectrometry. *Environmental Research*, 266:120494, 2025.

This paper by Nino-Suastegui et al. (2025), authored by a former student of my advisor working on non-targeted analysis of environmental contaminants, explores the use of liquid chromatography high-resolution mass spectrometry (LC-HRMS) combined with in silico tools like SIRIUS and CSI:FingerID for suspect screening and compound annotation in snow samples. The study successfully identified 17 known organic contaminants, mostly anthropogenic, including herbicides, insecticides, and plasticizers, and demonstrated how computational methods can aid in tentatively identifying unknown features beyond traditional library matching. The work emphasizes the challenge of numerous unidentified features remaining in environmental samples and shows that in silico analysis can complement existing databases, which is directly relevant to my research focus on improving computational techniques for molecule identification in complex environmental matrices.

- [8] Florent Olivon, Gwendal Grelier, Fanny Roussi, Marc Litaudon, and David Touboul. Mzmine 2 data-preprocessing to enhance molecular networking reliability. *Analytical Chemistry*, 89(15):7836–7840, 2017. PMID: 28644610.

This paper by Olivon et al. (2017) provides valuable insights into improving molecular networking workflows for untargeted mass spectrometry data analysis. It addresses limitations related to isomer resolution, quantification, and compound annotation by integrating MZmine 2 and custom Python scripts. Although the study focuses on natural products and plant extracts, its approach to preprocessing high-resolution MS/MS data can be adapted to my research on non-targeted analysis of organic contaminants in remote environments such as Mount Everest, Mt. Kilimanjaro, and Tibet River. The methodology’s emphasis on resolving isomers, incorporating retention time into clustering, and assigning molecular formulas aligns with my goal of computation-

ally identifying unknown molecules beyond simple mass-based annotation. This work demonstrates how rigorous data preprocessing combined with advanced clustering techniques can improve structural annotation accuracy. It is highly relevant for developing robust machine learning assisted molecular networking pipelines in environmental and analytical chemistry contexts.

- [9] Daniel Petras Louis-Félix Nothias Mingxun Wang Allegra T. Aron Annika Jagels Hiroshi Tsugawa et al. Schmid, Robin. Ion identity molecular networking for mass spectrometry-based metabolomics in the gnps environment. *Nature Communications*, 12(3832), 2021.

This research article on Ion Identity Molecular Networking (IIMN) significantly advances the field of non-targeted metabolomics by providing a robust computational framework for connecting different ion species of the same molecule within high-resolution mass spectrometry data. While much of the prior work in my project focuses on identifying unknown organic contaminants in remote environments such as Mount Everest, Mt. Kilimanjaro, and Tibet River—areas where many compounds remain uncharacterized—this paper offers valuable insights into how molecular networking can be used to cluster related ions and structurally similar molecules beyond simple mass-to-charge matching. The integration of ion identity networks into molecular networking enhances the ability to propagate annotations, expand spectral libraries, and improve the connectivity of unknown features. This ultimately facilitates more comprehensive and accurate characterization of complex environmental samples. I see this approach as a vital computational tool for my research, enabling me to better understand the relationships between unknown molecules and known clusters, and to develop a more effective non-targeted analysis framework grounded in chemistry and computational analysis.

- [10] Emma L. Schymanski, Junho Jeon, Rebekka Gulde, Kathrin Fenner, Matthias Ruff, Heinz P. Singer, and Juliane Hollender. Identifying small molecules via high resolution mass spectrometry: Communicating confidence. *Environmental Science & Technology*, 48(4):2097–2098, 2014.

This viewpoint article is essential for non-targeted analysis (NTA) research using HR-MS, such as the computational identification of unknown contaminants in remote environments. It proposes a five-level confidence system for molecular identification, addressing the limitations inherent when reference standards are absent. The framework provides the necessary rigor for reporting the results of computational tools like molecular networking or cluster analysis. Specifically, Level 3 (Tentative candidate(s)) helps classify structures where computational methods suggest isomers, while Levels 4 and 5 establish clear reporting standards for masses and formulas that serve as critical input or nodes in network-based analysis, enabling transparent data exchange and future traceability of unknown molecules.

- [11] Weiting Ye, Jingcheng Li, and Xianfa Cai. Mfgnn: Multi-scale feature-attentive graph neural networks for molecular property prediction. *Journal of Computational Chemistry*, 46(3):e70011, 2025.

This study introduces MfGNN, a multi-scale feature-attentive graph neural network designed for molecular property prediction by capturing both atom-level and fragment-level structural features. While its primary focus is on predicting molecular properties, the methodologies—particularly the integration of fragment-based representations and inter-fragment relationship modeling—offer valuable insights for non-targeted analysis in high-resolution mass spectrometry. Its ability to systematically incorporate substructural information and learn complex molecular relationships aligns with the objectives of molecular networking and cluster analysis, moving beyond traditional structure assignment based solely on mass spectra. Incorporating such graph-based machine learning frameworks can enhance the annotation, classification, and clustering of unknown molecules in environmental samples, especially in remote and contaminated regions. Therefore, this work provides a computationally advanced approach that can significantly contribute to the development of chemoinformatics tools for high-resolution mass spectrometry in non-targeted environmental studies.

- [12] Weimin Zhu, Yi Zhang, Duancheng Zhao, Jianrong Xu, and Ling Wang. Hignn: A hierarchical informative graph neural network for molecular property prediction equipped with feature-wise attention. *Journal of Chemical Information and Modeling*, 63(1):43–55, 2023. PMID: 36519623.

This paper on HiGNN offers valuable insights into leveraging advanced graph neural network architectures for molecular property prediction by incorporating hierarchical information and feature-wise attention mechanisms. While my research focuses on the computational identification of unknown molecules through high-resolution mass spectrometry in remote environments, the methodologies presented in this study are highly relevant for developing more sophisticated algorithms for non-targeted analysis. Specifically, the approach of molecular fragmentation and hierarchical representation can be adapted to analyze mass spectral data, enabling more accurate clustering and molecular networking of unknown organic contaminants. Integrating such deep learning models can enhance our capacity to classify and interpret complex datasets beyond traditional structure assignment, making them a promising tool in the non-targeted analysis of environmental samples in extreme and underexplored locations.